Algebra 1 – Module 6 Practice Test

Full Name: ______________________________

Period: _________

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Which graph shows the solution set of $x \le 6$?

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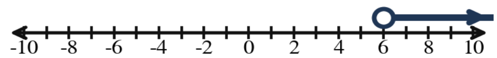

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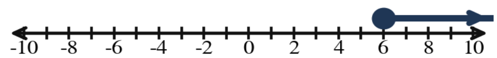

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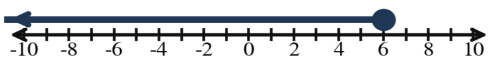

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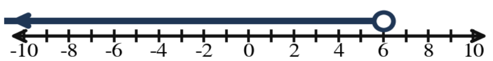
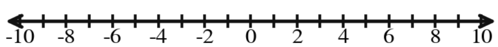
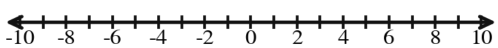


2. Solve and graph: $x + 8 < 2$



3. Solve and graph: $6y - 1 < 7y + 6$



4. Solve: $-3x + 7 \le -14$

5. Solve: $6(x - 4) > -18$

6. Solve: $7x - 6 \le 3x + 2$

7. Solve the compound inequality: $-7 < x + 2 \le -3$

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8. Solve the compound inequality: $x + 6 \le 4$ or $x - 3 \ge 5$

9. A rectangular banner has width 7 ft, and the ribbon border sewn around its edge can be at most 40 ft long. Write and solve an inequality to find the possible lengths.

10. Tomás has $18 saved and earns $14 each week walking dogs. He needs AT LEAST $116 for a guitar amplifier. Write and solve an inequality to find how many weeks he must save.

11. Error Analysis — Rex graphed $x < 5$ with a CLOSED circle at 5 and shading to the RIGHT. Identify the mistake.

12. A bakery's dough proofing box must stay between 78°F and 86°F, inclusive. Write a compound inequality for the allowed temperatures t.

13. Solve the compound inequality: $2 \le 5x - 8 < 27$

Cumulative Review — ACT Style. Circle the best answer.

14. Which is the equation of the line with slope 5 and y-intercept −1?

- Ⓐ $y = -x + 5$
- Ⓑ $y = 5x + 1$
- Ⓒ $y = 5x - 1$
- Ⓓ $y = x - 5$

15. What is the slope of the line through (−2, 5) and (4, −7)?

- Ⓔ −2
- Ⓕ 2
- Ⓖ −1/2
- Ⓗ 6

16. Let $f(x) = 8x - 5$. What is $f(4)$?

- Ⓐ 7
- Ⓑ 27
- Ⓒ 37
- Ⓓ −5